

Validating and Extending a Chapter-Aligned Moodle Engagement Metric Across STAT0002 and STAT0004

Evidence from online, hybrid and in-person Moodle delivery cohorts

Research Internship Report
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Abstract

Virtual learning environments such as Moodle record when students access course resources, how often they return, and which types of activities they use. These logs offer a scalable route to early student support, but simple click counts are hard to interpret and supervised predictive models typically require historical grade data. This report evaluates a theoretically grounded, chapter-aligned engagement metric that summarises three behaviours for each student: **Frequency** (the number of chapter-specific study sessions completed), **Immediacy** (how promptly a student first accesses newly released chapter material), and **Diversity** (the breadth of distinct resource types used within a chapter). Each component is min-max scaled relative to peers within the same chapter and teaching week, producing scores between 0 and 1, and the three are added to form a chapter-level score, $IDF = F + I + D \in [0, 3]$. Summing chapter scores across all material released by week t gives a cumulative engagement measure that can be updated in real time without outcome data. Building on the metric proposed by Johnston et al. (2025), this report applies the fixed measure to five cohorts from two UCL undergraduate statistics modules: STAT0002 (a first-year probability and statistics course) and STAT0004 (a second-year programming and practical statistics course), together comprising $N = 1,291$ students and spanning online, hybrid and in-person delivery periods not used in the original validation. The report asks whether the metric travels to new cohorts, which component carries the most distinct information about final grades, and whether the engagement–grade association is uniform across the grade distribution or concentrated among lower-performing students. Across the pooled cohorts, cumulative IDF shows a moderate positive association with final grade ($r = 0.32$). Once the shared variance among the three components is removed using a relative-importance decomposition, Diversity and Immediacy account for more unique explained variance than Frequency at week 11 (mean shares of 37%, 36% and 27%), even though all three correlate similarly with grade in bivariate analysis. The engagement–grade association is also stronger among lower-performing students: in every cohort, the slope at the 10th percentile of the grade distribution exceeds the median slope by an average of 2.3 grade points per standard deviation of engagement, and indicator-specific quantile models suggest that Immediacy is particularly important in the lower tail. Together, the findings support using the metric for low-cost, supportive monitoring rather than high-stakes prediction. Comparisons across delivery periods remain descriptive because delivery mode is confounded with cohort and academic year.

1 Introduction

Student engagement, the time and effort students devote to learning activities, is widely linked to academic performance, satisfaction and retention. In online and blended higher education, part of the *behavioural* dimension of engagement is recorded automatically by virtual learning environments (VLEs): which resources students access, how often they return, and how promptly

they engage with newly released material. Moodle logs therefore offer a scalable record of observed study behaviour, but they remain only a proxy. They cannot observe offline study, motivation, attendance or the quality of students’ understanding (Henrie et al., 2015).

This creates a practical measurement problem. Simple indicators such as total clicks are easy to compute but difficult to interpret pedagogically; a high count need not represent broad or sustained engagement (Fincham et al., 2019). Threshold systems can flag low activity, but their cut-offs are often arbitrary and may not generalise. Supervised predictive models can be more flexible, yet they require historical outcome data and can fail when moved to new modules, cohorts or delivery contexts (Gašević et al., 2016). For early student support, an attractive alternative is an interpretable metric that is grounded in engagement theory, can be updated during the term, and can be validated without assuming that a low log count is equivalent to low effort.

Johnston et al. (2025) propose such a metric by adapting the earlier course-wide engagement score of Chong and Wong (2019). The original course-wide score combines five indicators after the course has finished. Johnston et al. instead align the measurement with how teaching material is released: for each student, chapter and week, they compute Frequency, Immediacy and Diversity, scale each component relative to peers, and add them into a chapter-level IDF score. Frequency captures the volume of chapter-specific study sessions, Immediacy captures how soon a student first engages with a chapter after it becomes available, and Diversity captures the breadth of resource types accessed. Summing chapter scores over the term gives a cumulative engagement trajectory that can be monitored in real time.

The present report is a secondary validation and extension of that fixed metric, not a new metric-development exercise. It applies the chapter-aligned IDF measure to five cohorts from two UCL undergraduate statistics modules. STAT0002 is a first-year course introducing probability and statistical theory, represented here by an online delivery (2020–21) and an in-person delivery (2022–23). STAT0004 is a second-year programming and practical statistics course covering statistical computing and data analysis, and is the only module spanning all three delivery conditions: online (2020–21), hybrid (2021–22) and in-person (2022–23). These five cohorts were not used in the original validation, providing a genuinely independent test of the metric.

The analysis pursues two substantive questions. The first concerns component importance: once the strong correlations among Frequency, Immediacy and Diversity are accounted for, which component carries the most unique information about final grades? A relative-importance decomposition due to Lindeman et al. (1980), known as the LMG method, addresses this by averaging each predictor’s contribution to explained variance over all possible orderings in which the three components could enter a regression model. The second question asks whether the engagement–grade association is uniform across the grade distribution or stronger among students already at academic risk. Quantile regression estimates the slope between engagement and grade separately at different percentiles of the grade distribution, making it possible to ask whether low engagement is more consequential for lower-performing students than for the average student. Supporting analyses assess the overall pooled association, signal timing, engagement trajectories, early-warning precision, assessment-component alignment and robustness to analytic choices. Delivery-period comparisons are treated as descriptive boundary checks rather than causal tests because delivery mode is perfectly confounded with cohort and academic year.

2 Background and metric context

Online behavioural engagement refers to observable, sustained interaction with learning activities in a digital environment. Engagement theory distinguishes behavioural, emotional and cognitive forms of involvement; Moodle logs capture primarily the behavioural dimension, participation, effort and attention to academic tasks, rather than motivation or understanding (Fredricks et al., 2004; Henrie et al., 2015; Martin and Borup, 2022). It is only a *proxy* for learning: logs record

what happens inside the VLE, not offline study, in-person attendance or the quality of students’ reasoning. Even so, when behavioural engagement is measured coherently and validated against academic outcomes, it can support timely monitoring and targeted support.

The course-wide metric of Chong and Wong (2019) combines five indicators (Immediacy, Frequency, Diversity, Recency and Interval) into a single retrospective score aggregated across an entire course. That design is useful for end-of-term analysis but poorly suited to weekly monitoring, and it treats the course as homogeneous rather than aligned with its chapter structure. For real-time monitoring this is problematic: a student may be on schedule with newly released material while having little activity on later chapters that are not yet relevant, or may return frequently to old material without keeping pace with current content.

Johnston et al. (2025) address this by moving from a course-wide score to a chapter-aligned score. At each week t , and for each released chapter k , three raw indicators are calculated from chapter-specific study sessions: **Frequency**, the number of study sessions associated with the chapter; **Immediacy**, the delay between the chapter’s first observed availability and the student’s first chapter-specific session, transformed so earlier access receives a higher score; and **Diversity**, the number of distinct Moodle activity or resource types accessed within that chapter. Each raw indicator is min-max scaled across all students for the same chapter k and week t , placing it on a common $[0, 1]$ scale relative to peers. For example, the scaled Immediacy indicator is

$$I_{k,t}^{(i)} = \frac{\tilde{I}_{k,t}^{(i)} - \min_i \tilde{I}_{k,t}^{(i)}}{\max_i \tilde{I}_{k,t}^{(i)} - \min_i \tilde{I}_{k,t}^{(i)}},$$

and the same transformation is applied to $\tilde{F}$ and $\tilde{D}$. A student who scores 1 on all three components engaged with the chapter earlier, more frequently, and across a broader range of resource types than any of their peers. The chapter-level IDF score is then the simple sum

$$IDF_{k,t}^{(i)} = F_{k,t}^{(i)} + I_{k,t}^{(i)} + D_{k,t}^{(i)} \in [0, 3],$$

so a high score means that a student engages often, early and broadly with that chapter. Chapter scores are then summed over released chapters to obtain an overall cumulative engagement measure for student i at week t . Recency and Interval are excluded from the real-time version because they become unstable when measured before the course has finished.

Chapter alignment matters because engagement is most interpretable when measured in step with how material is released. For early-warning use, practitioners also need to understand the recall–precision trade-off: flagging the lowest-engagement students may capture many who ultimately struggle, but also many who would have succeeded without intervention (Conijn et al., 2017). The present report takes the metric definition as given and focuses on how it behaves on new UCL cohorts and what additional diagnostic insight can be extracted from its components and related temporal patterns.

3 Data and engagement measures

The analysis uses five module-year deliveries from two undergraduate statistics modules at University College London. STAT0002 is a first-year foundations module represented here by an online cohort (2020–21) and an in-person cohort (2022–23). STAT0004 is a second-year module and the only module in this dataset spanning all three delivery conditions: online (2020–21), hybrid (2021–22) and in-person (2022–23). Each delivery covers eleven teaching weeks with up to ten labelled chapters. Final grades are recorded on a 0–100 scale; grades below 50 were treated as low performance, and grades below 40 as failure. Students with a recorded grade of zero, typically reflecting exam absence, were excluded.

Table 1 summarises the cohorts. The main engagement–grade sample comprises $N = 1,291$ students with both engagement records and valid final grades after exclusions. The latent trajectory analysis below uses $N = 1,290$ students with complete weekly engagement profiles suitable for class assignment; the difference of one student reflects missing weekly coverage in the trajectory model, not an error in the cohort inventory.

Table 1: Cohort inventory for the main engagement–grade sample ($N = 1,291$).

Module	Year	Condition	N	Mean (SD)	% < 50	% < 40
STAT0002	2020–21	Online	342	66.8 (13.8)	9.9	1.8
STAT0002	2022–23	In-person	268	67.4 (13.0)	10.4	2.6
STAT0004	2020–21	Online	312	62.7 (12.2)	12.5	2.9
STAT0004	2021–22	Hybrid	172	71.9 (8.8)	1.7	0.0
STAT0004	2022–23	In-person	197	70.6 (11.3)	3.6	1.0
Total			1,291			

Source: author’s analysis of UCL Moodle log data.

Two design features constrain interpretation. First, the share of low-performing students varies widely across cohorts (from 1.7% to 12.5%), which limits how sharply any early-warning flag can perform in the highest-achieving deliveries. Second, delivery condition is **perfectly confounded** with academic year and cohort: online corresponds to 2020–21, hybrid to 2021–22, and in-person to 2022–23. Comparisons across conditions therefore describe different module-year contexts rather than randomly assigned delivery modes, and no causal claim about online, hybrid or in-person delivery can be supported.

The modules also differed in assessment structure. STAT0002 in-person records separate exam, participation and peer-marking components; STAT0004 deliveries include coursework and quiz marks, with a group component in the in-person year. This matters because Moodle engagement may align more closely with individual or exam-style assessment than with group, peer-marked or participation components. Where component marks were available, this report therefore analyses whether the engagement metric correlates more strongly with individually assessed work than with collaborative or peer-assessed components.

The two modules also differ in the strength of the engagement–grade link across deliveries. STAT0002 cohorts show end-of-term correlations of $r = 0.28$ – 0.41 , whereas STAT0004 ranges from $r = 0.19$ to $r = 0.30$. As interpretive context, the two modules may differ in ways that affect how well Moodle activity proxies study behaviour; several factors may contribute, though they cannot be disentangled in this observational design. STAT0002 is a first-year foundations module with substantial individually assessed exam weight, where sustained access to lecture materials and problem sheets may track preparation more directly. STAT0004 is a second-year module with greater student autonomy, more coursework and group-assessment weighting, and possibly more study away from Moodle. Each of these factors would weaken the observable link between VLE logs and final outcomes. How tightly Moodle resources are organised by chapter may also matter: where chapter labelling aligns less clearly with assessed work, a chapter-aligned metric would legitimately carry less outcome signal, as Johnston et al. (2025) report for less structured courses.

3.1 Engagement feature construction

The supplied engagement table contains weekly chapter-level contribution columns for Frequency, Immediacy and Diversity. These are the operational form of the Johnston et al. IDF components used in this report: each column corresponds to a student–week–chapter contribution for one component. Before modelling, the data were checked to clarify their time semantics. A recon-

struction from the raw Moodle event log showed that the supplied contribution values behave as *per-week increments*, not as already cumulative totals. This matters because the paper’s real-time metric is cumulative up to week t ; therefore the analysis must first build weekly totals and then accumulate them explicitly.

For each student and week, contribution values were summed across released chapters to obtain weekly Frequency, Immediacy and Diversity totals. The weekly composite was then defined additively as

$$\text{IDF}_{\text{week}} = \text{freq}_{\text{week}} + \text{imm}_{\text{week}} + \text{div}_{\text{week}},$$

and cumulative measures were obtained by summing across weeks: separately for Frequency, Immediacy, Diversity and the composite IDF:

$$\text{cum_IDF}_{i,t} = \sum_{s=1}^t \text{IDF}_{\text{week},i,s}.$$

This additive definition follows the chapter-level formulation of Johnston et al. (2025); it is not a multiplicative composite and no outcome data are used to create the engagement measures. The reconstruction check confirmed that the supplied contributions are most consistent with the intended per-week Frequency and Diversity definitions, while Immediacy is retained as supplied because it depends on first-access timing and release-date assumptions.

For extension analyses, additional timing and regularity features were derived from event logs (for example, variation in weekly activity and gaps between sessions). Latent-class mixed models were also fitted to normalised weekly cumulative IDF curves to identify distinct engagement trajectories over the term.

4 Methods

4.1 Analytical strategy

The analysis proceeds in four stages. First, the supplied weekly engagement contributions are checked against the Moodle event log and converted into a complete student-by-week panel. Second, the cumulative IDF measure is related to final grades within each cohort and then synthesised across cohorts. Third, the composite is decomposed into Frequency, Immediacy and Diversity, and the engagement–grade association is examined across the grade distribution rather than only at its average. Finally, temporal patterns, assessment-component alignment, early-warning performance and a set of robustness checks assess how the metric might be interpreted and used in practice. All analyses are associational: final grades are outcomes used to evaluate construct validity, not targets used to construct the engagement score.

4.2 Analytical sample and feature preparation

The initial dataset contains one or more weekly records for each student who appears in the supplied engagement table. Final grades were linked by the anonymised student identifier within cohort. Students with a recorded final grade of zero were removed before analysis because, in these data, a zero normally indicates examination absence rather than achieved performance. The resulting analysis sample contains students with both usable Moodle engagement records and $0 < \text{final grade} \leq 100$.

The engagement export is *ragged*: a student may have no row for a week in which they did not engage, and chapter columns are missing until a chapter is released. Treating an absent student-week as missing data would remove inactive students from weekly summaries and bias the sample towards active users. The analysis therefore completes the grid of every included student

across teaching weeks 1–11. A missing student-week is coded as zero observed engagement, while unreleased chapter values remain structurally unavailable. This distinction allows silent students to remain in the cohort denominator and ensures that a cumulative score carries forward unchanged when no new activity occurs.

4.3 Verification and construction of weekly engagement

The supplied `sem` table provides chapter-level contributions (`freq_contrib`, `imm_contrib` and `div_contrib`), but does not itself document whether each value represents activity in the current week or activity accumulated up to that week. This distinction is critical: using a cumulative contribution as if it were a weekly increment would count the same study behaviour repeatedly when constructing a running total.

To resolve this, the verification procedure reconstructs raw Frequency and Diversity from the event log for the hybrid STAT0004 2021–22 cohort. Frequency is the number of distinct chapter-labelled study sessions; Diversity is the number of distinct resource types accessed in chapter-labelled sessions. Each quantity is reconstructed twice, once from all activity up to week t and once from activity in week t only. For every available chapter-week, students with no reconstructed activity receive zero. The rank ordering of each reconstructed quantity is then compared with the ordering of the supplied contribution using Spearman’s ρ . Spearman correlation is appropriate because min–max scaling is monotonic within a chapter-week: if the raw quantity and the supplied contribution represent the same underlying behaviour, their student ranks should agree even when their numerical scales differ. The per-week reconstruction produced the substantially stronger agreement for both Frequency and Diversity, so the supplied contributions are treated as per-week increments. Immediacy is retained as supplied because its reconstruction requires the original release date and first-access conventions.

For student i in week t , chapter contributions are summed over released chapters to create weekly component totals:

$$F_{i,t}^{\text{week}} = \sum_k F_{i,k,t}, \quad I_{i,t}^{\text{week}} = \sum_k I_{i,k,t}, \quad D_{i,t}^{\text{week}} = \sum_k D_{i,k,t}.$$

The weekly composite is their additive sum,

$$\text{IDF}_{i,t}^{\text{week}} = F_{i,t}^{\text{week}} + I_{i,t}^{\text{week}} + D_{i,t}^{\text{week}}.$$

The cumulative measure available at week t is then

$$\text{cumIDF}_{i,t} = \sum_{s=1}^t \text{IDF}_{i,s}^{\text{week}},$$

with analogous cumulative totals for Frequency, Immediacy and Diversity. This construction reproduces the practical purpose of the chapter-aligned metric: at any week, the score uses only activity observed by that point in the course.

4.4 Standardisation and outcome definitions

Raw cumulative scores naturally increase through the term as students have more opportunity to engage and more chapters are released. Cohorts also differ in class size, delivery context, resource design and the distribution of Moodle activity. For regression analyses, each weekly and cumulative engagement measure is therefore standardised within each cohort and week:

$$z_{i,t} = \frac{x_{i,t} - \bar{x}_{c,t}}{s_{c,t}},$$

where $\bar{x}_{c,t}$ and $s_{c,t}$ are the mean and standard deviation among students in cohort c at week t . A coefficient for z_{cumIDF} can therefore be read as the expected grade difference associated with one standard deviation higher engagement relative to peers in the same cohort at the same stage of term. Raw values are retained for rank correlations and threshold-based early-warning analyses, where within-cohort ordering is the quantity of interest. Final grade is analysed continuously on the 0–100 scale. For early-warning analyses, a grade below 50 is defined as low performance and a grade below 40 as failure.

4.5 Overall association and pooled synthesis

Within every cohort and week, Spearman’s rank correlation was calculated between cumulative engagement and final grade. This non-parametric statistic tests whether students with relatively higher engagement also tend to rank higher on final grade, without assuming a linear relationship or normally distributed scores. The week-by-week correlations show how the grade association evolves as information accumulates.

Because a single pooled correlation would be dominated by the largest cohort and would assume the same association in every delivery, cohort-specific correlations were combined using a random-effects meta-analysis implemented in *metafor* (Viechtbauer, 2010). Fisher-transformed correlations were pooled and then converted back to the correlation scale; the reported I^2 statistic quantifies the proportion of observed between-cohort variation not attributable to sampling error. A mixed-effects regression of final grade on standardised cumulative IDF, with cohort-specific slopes, complements this rank-based analysis by expressing the association as grade points per standard deviation of engagement.

4.6 Component importance

Frequency, Immediacy and Diversity measure related aspects of the same underlying study behaviour, and are consequently strongly correlated in these data. Ordinary multiple-regression coefficients are not a reliable answer to the question “which component matters most?” under such collinearity: their size and even their sign can change according to the order in which correlated predictors are entered. The analysis therefore uses the Lindeman–Merenda–Gold (LMG) relative-importance method (Lindeman et al., 1980).

For each cohort at weeks 6 and 11, an ordinary least-squares model predicts final grade from the three standardised cumulative components. With three predictors, there are $3! = 6$ possible orders in which they could be entered. LMG calculates the additional R^2 attributable to each component for every order and averages these increments. The three resulting shares sum to the model’s total R^2 and do not privilege an arbitrary predictor order. Week 6 represents a plausible mid-course support point, while week 11 represents the complete teaching period. Bivariate Spearman correlations are also pooled for each component; comparing these with the LMG shares distinguishes simple association with grade from unique explanatory contribution after overlap among components is considered.

4.7 Association across the grade distribution

Mean-based models can obscure whether the engagement–grade association is similar for students at different achievement levels. Quantile regression (Koenker and Bassett, 1978) estimates the association at selected conditional quantiles:

$$Q_{\tau}(\text{grade}_i \mid z_{\text{cumIDF},i}) = \beta_0(\tau) + \beta_1(\tau)z_{\text{cumIDF},i},$$

where $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$. The coefficient $\beta_1(\tau)$ is the grade-point difference associated with a one-standard-deviation increase in cumulative engagement at quantile τ . The contrast

$\beta_1(0.1) - \beta_1(0.5)$ tests descriptively whether the lower-tail slope exceeds the median slope. It is accompanied by a 95% student-level bootstrap interval based on 500 resamples within cohort. Separate univariate models for Frequency, Immediacy and Diversity identify whether any component is especially salient in the lower tail. All component and quantile analyses are run at weeks 6 and 11 to contrast mid-course and end-of-term patterns.

4.8 Timing, early warning and operational performance

To distinguish an early stable signal from a merely early but weak signal, the analysis estimates a stabilisation week separately for each cohort. Students are resampled with replacement 2,000 times. For each resample and week w , the difference $\rho_{11} - \rho_w$ is calculated, where ρ_w is the Spearman correlation between week- w cumulative IDF and final grade. The first week for which the 95% bootstrap interval for this difference includes zero is called the stabilisation week: at that point, the observed association is statistically indistinguishable from the week-11 association. This is not a claim that the signal is strong at that week; the magnitude of ρ_w is reported separately.

For a simple early-warning rule, students in the bottom 20% of cumulative IDF within each cohort are flagged. The rule is evaluated against low performance (grade < 50) using three complementary quantities:

$$\text{recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}, \quad \text{precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}.$$

Recall answers how many eventual low performers are flagged; precision answers how many flagged students ultimately have low grades. Area under the ROC curve (AUC) measures the probability that a randomly selected low performer has lower engagement than a randomly selected non-low performer. AUC, recall and precision are reported together because low-performance prevalence varies across cohorts and AUC alone can conceal a high false-positive burden (Conijn et al., 2017; Johnston et al., 2025).

4.9 Assessment validity, temporal features and trajectories

Where separate assessment-component marks are available, cumulative week-11 IDF is correlated with each component. Steiger’s dependent-correlation test (Steiger, 1980) then tests whether the correlation with an individual exam or quiz is larger than the correlation with a peer-marked or group component, using the fact that both correlations were calculated from the same students. This analysis assesses whether Moodle engagement aligns most closely with outcomes that reflect individual study behaviour rather than collaborative assessment (Almond, 2009).

The core F/I/D metric describes volume, promptness and breadth, but does not fully describe *how regularly* students study. At week 6, six additional event-log features are calculated: activity evenness across weeks, variability in gaps between active days, the share of late-night activity, breadth of Moodle components used, a capped time-on-task proxy based on inter-event gaps, and the share of events occurring more than two weeks after a chapter first appears. Nested linear models compare R^2 from the F/I/D-only model with R^2 after adding these features. The nested-model F test asks whether the temporal features jointly explain grade variation beyond the three core components.

Finally, weekly within-cohort standardised IDF profiles are modelled using latent-class mixed models in the `lcmm` framework (Proust-Lima et al., 2017). Models with one to three quadratic trajectory classes are compared using the Bayesian information criterion (BIC), which balances fit against the additional complexity of more classes. The selected classes are labelled from their estimated mean profiles and compared on final grade using a Kruskal–Wallis test. To examine early operational usefulness, a student’s partial profile through week w is assigned to

the closest end-of-term class centroid; agreement, recall and precision for the low-steady class are then reported across weeks 3–9.

4.10 Robustness checks

Robustness analyses address specific alternative explanations rather than creating additional headline tests. First, degree-programme adjustment tests whether the Diversity association could reflect programme composition in cohorts where programme codes are available. Second, an extenuating-circumstances sensitivity analysis recomputes correlations after excluding students with recorded extenuating-circumstance minutes. Third, the Benjamini–Hochberg procedure controls the false-discovery rate across the family of weekly engagement–grade correlations. Finally, early-warning estimates are recalculated using bottom 10%, 20% and 30% thresholds and compared for cumulative versus single-week engagement. A conclusion is considered robust when its direction and practical interpretation do not depend on any one of these choices.

5 Results

Results are reported first for the overall association between cumulative engagement and final grade, then for the component and grade-distribution analyses, before turning to timing, trajectories, assessment alignment, early-warning performance and robustness.

5.1 Overall engagement–grade association

Across all five cohorts, cumulative engagement is positively and moderately associated with final grade. The random-effects pooled rank correlation for cumulative IDF at week 11 is $r = 0.32$ (95% CI [0.25, 0.38]), with low-to-moderate heterogeneity across cohorts ($I^2 = 39\%$). Frequency, Immediacy and Diversity show similar pooled associations when analysed separately (Table 2). On the grade scale, the mixed-effects estimate corresponds to approximately 3.8 grade points per one standard deviation increase in engagement. Figure 1 shows that every cohort lies above zero, but the association is far from perfect prediction: engagement is a useful but imperfect proxy for academic performance.

Table 2: Pooled rank correlations and mixed-effects slopes at week 11.

Indicator	Pooled r [95% CI]	I^2	Slope (SE)	Moderator p
Frequency	0.28 [0.22, 0.34]	27%	3.36 (0.53)	0.92
Immediacy	0.30 [0.24, 0.37]	37%	3.63 (0.58)	0.88
Diversity	0.31 [0.24, 0.37]	33%	3.62 (0.53)	0.87
IDF	0.32 [0.25, 0.38]	39%	3.79 (0.58)	0.89

Source: author’s analysis of UCL Moodle log data.

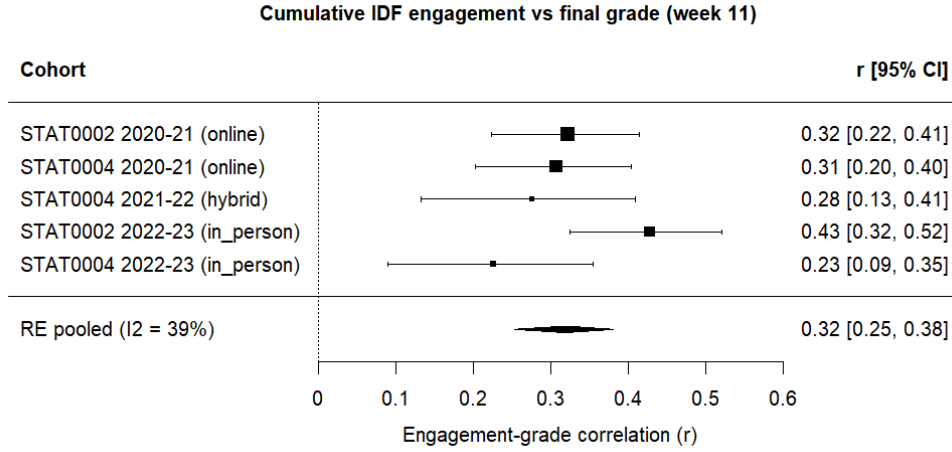


Fig. 1: Cohort-specific and pooled week-11 correlations between cumulative IDF engagement and final grade. All cohorts show positive associations; the random-effects estimate summarises the overall moderate engagement–grade relationship.

STAT0002 deliveries show stronger cohort-level associations than STAT0004 ($r = 0.28$ – 0.41 versus $r = 0.19$ – 0.30), consistent with the module-level patterns noted in Section 3. The pooled estimate nonetheless remains positive and moderate because every cohort contributes in the same direction.

When students are ranked by cumulative engagement and grouped into quintiles, final grades tend to rise across successive quintiles, particularly in STAT0002, where quintile separation is clearest (Figure 2). At weeks 3 and 6, the lowest-engagement quintile is the group whose grade distribution most often approaches or crosses the 50% low-performance threshold, while the highest-engagement quintile consistently shows the strongest outcomes. Separation between adjacent middle quintiles is weaker, which is consistent with a monotone but noisy relationship rather than sharp boundaries between moderate engagement levels. Together, the pooled correlation, positive cohort-level associations and quintile patterns support the conclusion that the metric *travels* to these new module-year contexts with a moderate, robust association.

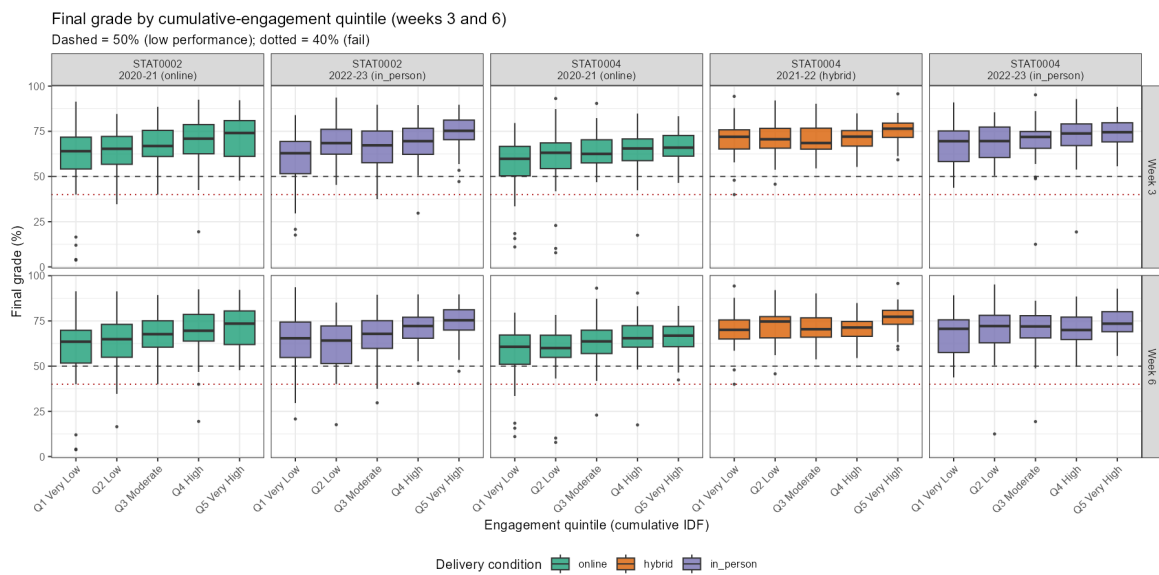


Fig. 2: Final-grade distributions by engagement quintile at weeks 3 and 6.

5.2 Component importance

Treating engagement as a single composite score confirms that it relates to grades, but obscures *which* learning behaviours carry unique predictive information. Frequency (how often students interact), Diversity (the variety of resource types accessed) and Immediacy (how promptly students engage with newly released content) are strongly correlated within cohorts: mean pairwise $|r| \approx 0.75\text{--}0.83$, with variance inflation factors between 2.6 and 4.4 at weeks 6 and 11. Under such collinearity, ordinary multiple-regression coefficients are unstable and can even change sign across cohorts; a method that partitions explained variance fairly is required.

Bivariate contrast. Bivariate pooled rank correlations at week 11 show similar associations for all three indicators when analysed separately: $r = 0.28$ (Frequency), $r = 0.30$ (Immediacy) and $r = 0.31$ (Diversity; Table 2). Taken alone, this would suggest that all components matter equally. The LMG results show that this is misleading once shared variance is removed.

End-of-term results. At end-of-term (week 11), Diversity accounts for the largest mean share of explained variance (37%), Immediacy is intermediate (36%), and Frequency the smallest (27%). Frequency has the smallest LMG share in every cohort (Table 3, Figure 3). Cohort-level patterns vary: Immediacy dominates in STAT0002 online (45%), whereas Diversity dominates in STAT0004 online and in-person (45% and 42% respectively).

Table 3: LMG relative-importance shares (%) at week 11.

Cohort	Frequency	Immediacy	Diversity
STAT0002 online	28	45	27
STAT0002 in-person	28	36	36
STAT0004 online	23	32	45
STAT0004 hybrid	31	32	38
STAT0004 in-person	25	33	42
Mean	27	36	37

Source: author’s analysis of UCL Moodle log data.

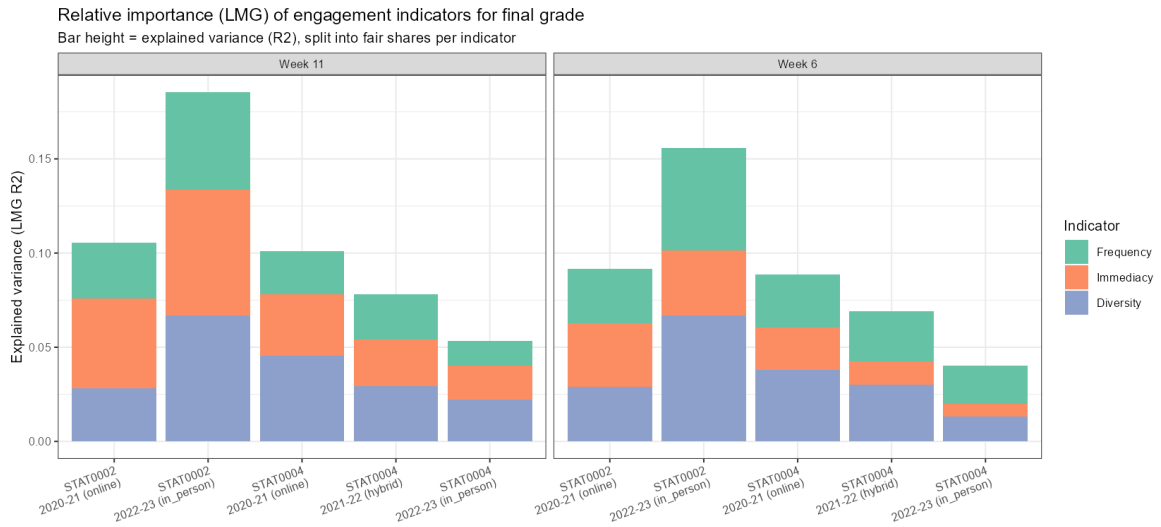


Fig. 3: LMG relative-importance shares for Frequency, Immediacy and Diversity by cohort at weeks 6 (left) and 11 (right). Diversity and Immediacy typically account for more unique explained variance than Frequency.

Mid-term results. The same decomposition at week 6, the course midpoint and a common decision point for early support, shows a broadly similar ranking: Diversity remains the top contributor on average (39%), with Immediacy lowest (24%) and Frequency intermediate (37%). One cohort (STAT0004 in-person) assigns a larger week 6 share to Frequency (50%), but Frequency never ranks first at week 11 in any cohort. The pooled pattern supports interpreting breadth of resource use as the most consistent unique signal, with promptness also carrying substantial information, especially in first-year STAT0002.

Table 4: LMG relative-importance shares (%) at week 6.

Cohort	Frequency	Immediacy	Diversity
STAT0002 online	31	37	32
STAT0002 in-person	35	22	43
STAT0004 online	32	25	43
STAT0004 hybrid	38	18	43
STAT0004 in-person	50	17	33
Mean	37	24	39

Source: author’s analysis of UCL Moodle log data.

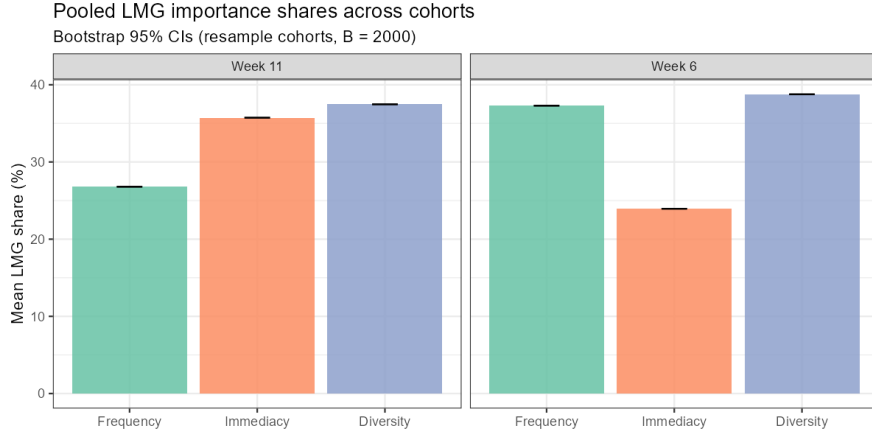


Fig. 4: Pooled mean LMG importance shares across cohorts at weeks 6 and 11.

5.3 Variation across the grade distribution

A single average association, whether from OLS or Spearman correlation, can hide whether engagement is more consequential for students already at academic risk. Quantile regression addresses this by estimating separate slopes at different points in the grade distribution.

Composite engagement. At week 11, the IDF slope is steeper at $\tau = 0.1$ than at $\tau = 0.5$ in **every cohort** (Table 5). On average, the lower-tail slope exceeds the median slope by 2.3 grade points per standard deviation of engagement. Slopes generally decrease from $\tau = 0.1$ through $\tau = 0.5$ to $\tau = 0.9$, indicating that engagement variation is most consequential among lower-performing students. For example, in STAT0002 in-person the slope is 7.9 points at $\tau = 0.1$ versus 5.0 at the median and 3.0 at $\tau = 0.9$. STAT0004 hybrid shows the smallest tail excess (0.6 points), suggesting cohort-specific heterogeneity in how sharply the lower tail diverges.

Student-level bootstrap confidence intervals for $\beta_1(0.1) - \beta_1(0.5)$ confirm the pattern in two of five cohorts at week 11 (STAT0002 and STAT0004 in-person); in the remaining cohorts the point estimates are positive but intervals are wide (Table 6). At week 6, three cohorts show positive descriptive contrasts and one (STAT0004 in-person) has a bootstrap interval excluding zero.

Table 5: Quantile-regression slopes for cumulative IDF at week 11 (grade points per +1 SD).

Cohort	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau_{0.1} - \tau_{0.5}$
STAT0002 online	5.32	3.22	2.58	2.11
STAT0002 in-person	7.88	5.04	2.96	2.84
STAT0004 online	6.06	3.15	2.17	2.91
STAT0004 hybrid	3.26	2.66	1.34	0.59
STAT0004 in-person	3.97	1.10	1.28	2.88

Source: author's analysis of UCL Moodle log data.

Table 6: Bootstrap 95% CI for $\tau_{0.1} - \tau_{0.5}$ contrast (week 11).

Cohort	Contrast	95% CI	CI > 0?
STAT0002 online	2.11	[-1.60, 4.79]	No
STAT0002 in-person	2.84	[0.16, 5.54]	Yes
STAT0004 online	2.91	[-0.85, 4.77]	No
STAT0004 hybrid	0.59	[-2.62, 4.49]	No
STAT0004 in-person	2.88	[0.26, 5.77]	Yes

Source: author's analysis of UCL Moodle log data.

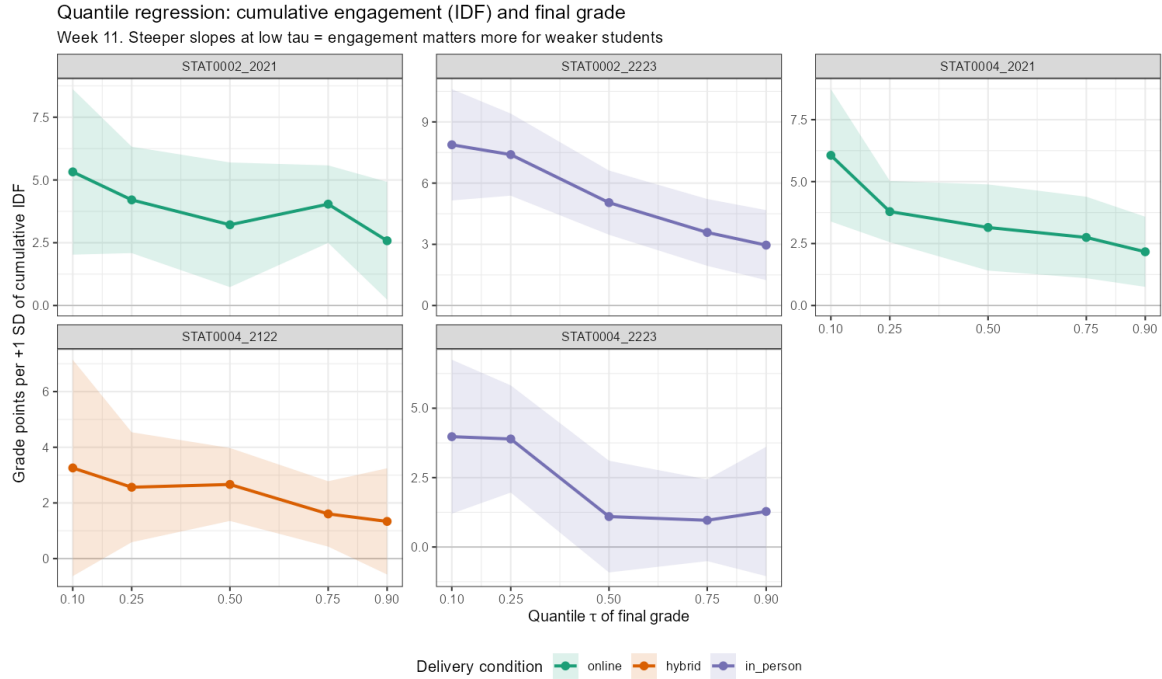


Fig. 5: Quantile-regression slopes of final grade on cumulative IDF at week 11, with 95% confidence intervals, by cohort.

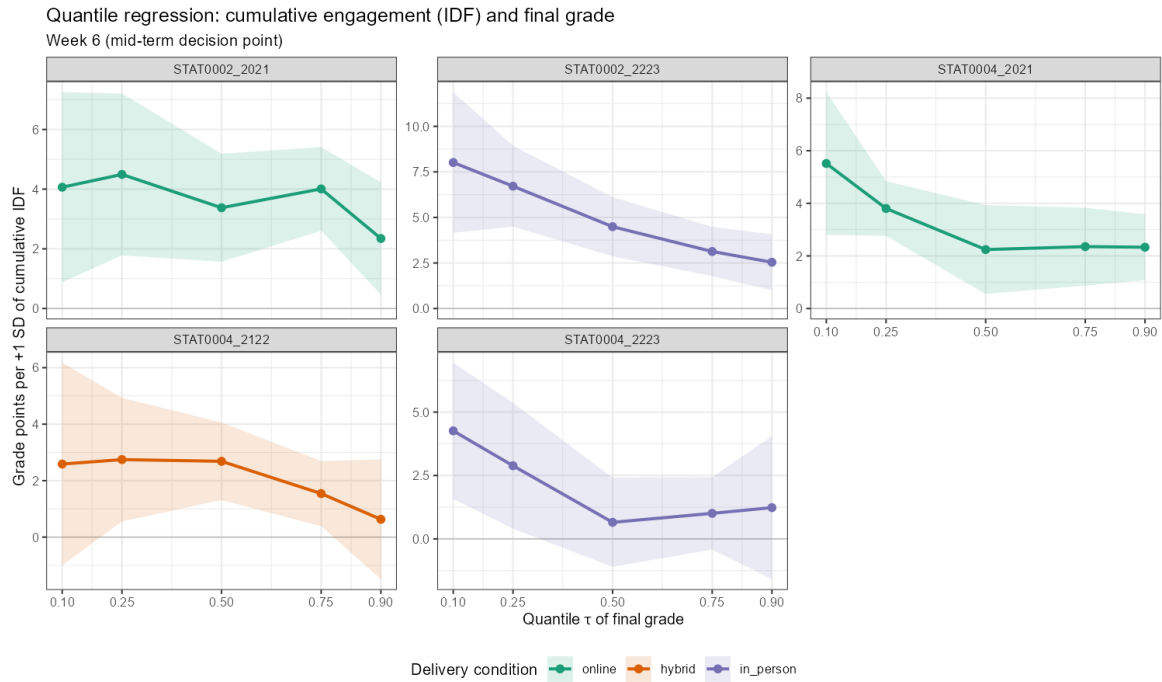


Fig. 6: Quantile-regression slopes at week 6 (mid-term). The lower-tail pattern is visible in most cohorts by the course midpoint.

Indicator-specific associations. Univariate quantile regression (one indicator at a time) shows that **Immediacy** has the largest mean slope at $\tau = 0.1$ across cohorts at week 11 (4.9 grade points per SD), followed by Diversity and Frequency (Figure 7). This complements the LMG finding that Diversity carries the most unique variance in multivariate models: promptness appears especially salient for distinguishing outcomes among lower-performing students when indicators are considered separately.

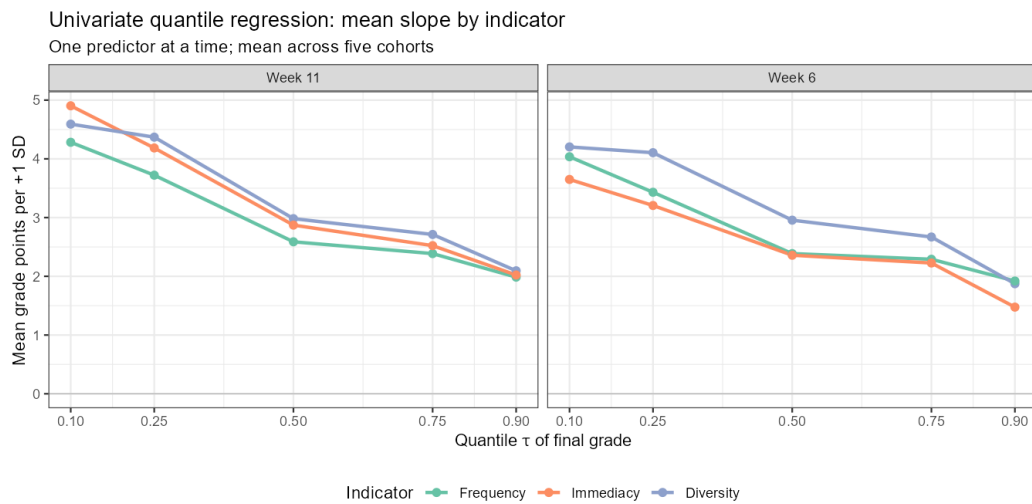


Fig. 7: Mean univariate quantile-regression slopes by indicator (weeks 6 and 11; mean across five cohorts). Larger coefficients at low τ indicate stronger associations among lower-performing students.

5.4 Timing and stabilisation

Bootstrap stabilisation analysis asks whether additional weeks materially change the engagement–grade correlation, not whether the correlation is already strong (Table 7, Figure 8). In four cohorts, the weekly correlation is statistically close to its end-of-term value from week 1 or 2, but this sometimes reflects a relatively flat and modest signal rather than impressive early predictive power. For example, STAT0004 in-person stabilises early yet has a weak end-of-term association ($r = 0.19$); STAT0004 hybrid similarly stabilises at week 1 with $r = 0.23$. By contrast, STAT0002 in-person shows a strengthening association that does not stabilise until around week 9, reaching the strongest cohort-level correlation ($r \approx 0.41$), suggesting that later engagement adds meaningful information in that delivery. Stabilisation timing and association strength should therefore be read separately: early stabilisation does not guarantee a strong early-warning signal.

Table 7: Stabilisation week and end-of-term IDF–grade correlation.

Cohort	Stabilisation week	End-of-term r
STAT0002 online	2	0.28
STAT0002 in-person	9	0.41
STAT0004 online	1	0.30
STAT0004 hybrid	1	0.23
STAT0004 in-person	1	0.19

Source: author’s analysis of UCL Moodle log data.

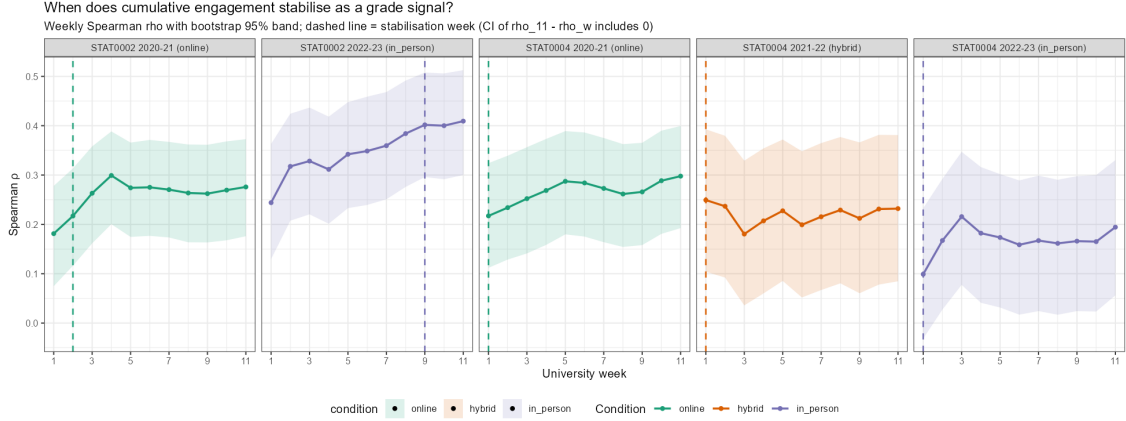


Fig. 8: Weekly Spearman correlation between cumulative engagement and final grade by cohort, with bootstrap stabilisation week marked. Early stabilisation can reflect a flat, modest signal as well as a strong one; compare stabilisation week with end-of-term r in Table 7.

5.5 Engagement trajectories

Latent-class analysis identifies three weekly cumulative IDF trajectory types: low-steady, high-steady and mid-variable (Figure 9). Final grades differ markedly across classes (Kruskal–Wallis $p = 2.6 \times 10^{-17}$). The high-steady class averages 72.9, compared with 63.2 for the low-steady class: a gap of about 9.7 grade points. The low-steady class also contains the highest share of low performers (15.7%). Sustained engagement *patterns* over the term therefore separate outcomes more sharply than a single cumulative score at one time point, and identify a group whose profile is persistently low rather than temporarily quiet.

Table 8: Trajectory classes and grade outcomes ($N = 1,290$).

Class	N	Mean grade	% < 50
Low-steady	439	63.2	15.7
High-steady	165	72.9	2.4
Mid-variable	686	68.3	5.5

Source: author’s analysis of UCL Moodle log data.

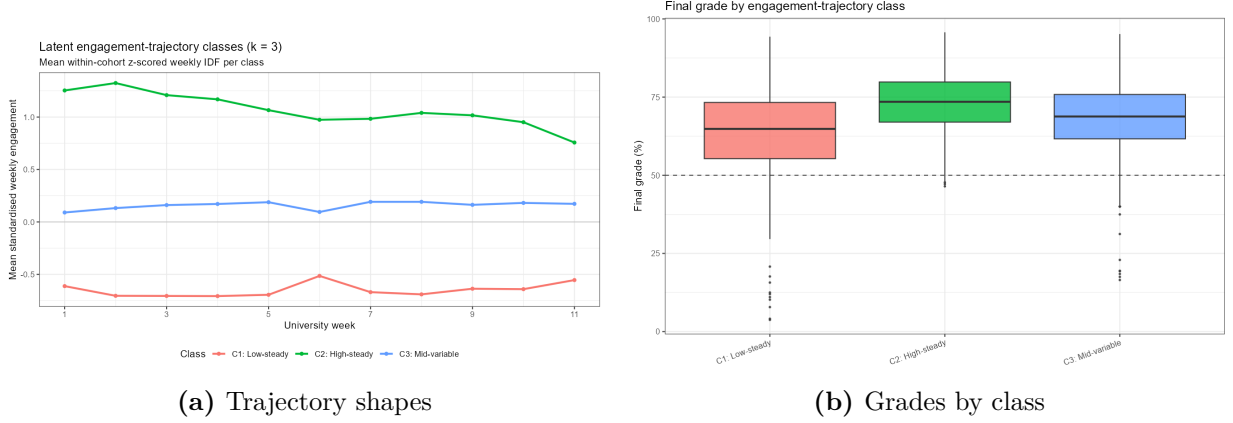


Fig. 9: Latent engagement trajectories (left) and final-grade distributions by trajectory class (right). Three trajectory types separate outcomes by roughly 9.7 grade points between the highest- and lowest-engagement profiles.

A natural operational question is how early a student can be assigned to a trajectory class with reasonable confidence. End-of-term class labels were therefore compared with early assignments made by matching each student’s partial weekly profile (weeks 1– w) to the corresponding segment of the end-of-term class centroids (Table 9, Figure 10). Overall agreement rises from 71% at week 3 to 90% by week 9. For the low-steady class specifically, recall reaches 81% by week 4 and 88% by week 6; precision that an early low-steady flag corresponds to the true end-of-term low-steady class is 77% at week 4 and 81% at week 6. These levels are materially higher than the bottom-quintile flag’s precision for identifying students who ultimately score below 50% (about 15–26%). However, among students flagged as low-steady early, only about 15% still score below 50%, similar to the quintile rule. Trajectory class is therefore a more reliable early label for *sustained low engagement* than the quintile is for *eventual failure*: operationally useful for targeted, pattern-based outreach from mid-term onwards, but not a sharp failure predictor.

Table 9: Early recovery of end-of-term trajectory class from partial weekly profiles.

Week	Agreement (%)	Low-steady recall (%)	Low-steady precision (%)	% flagged < 50
3	70.9	78.6	69.0	14.8
4	76.9	81.3	76.8	14.6
6	81.6	88.2	80.8	14.8

Source: author’s analysis of UCL Moodle log data.

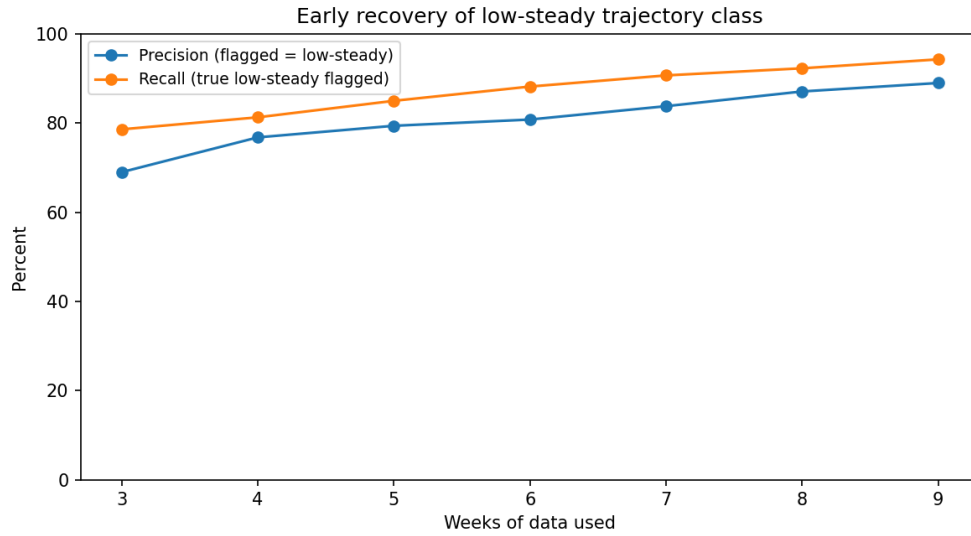


Fig. 10: Recall and precision for early assignment to the low-steady trajectory class as more weekly data accumulate. Agreement improves steadily from week 3 onwards.

5.6 Assessment alignment and temporal regularity

Engagement correlates more strongly with individually assessed components than with group- or peer-marked work where both are available (Figure 11). In STAT0002 in-person, the correlation with the exam grade ($r = 0.40$) is significantly larger than with peer-marking ($r = 0.20$; Steiger $z = 2.98$, $p = 0.003$). Other component contrasts point in the same direction but are not statistically significant. This supports the interpretation that Moodle engagement reflects individual study behaviour more faithfully than outcomes shaped by group dynamics or peer assessment.

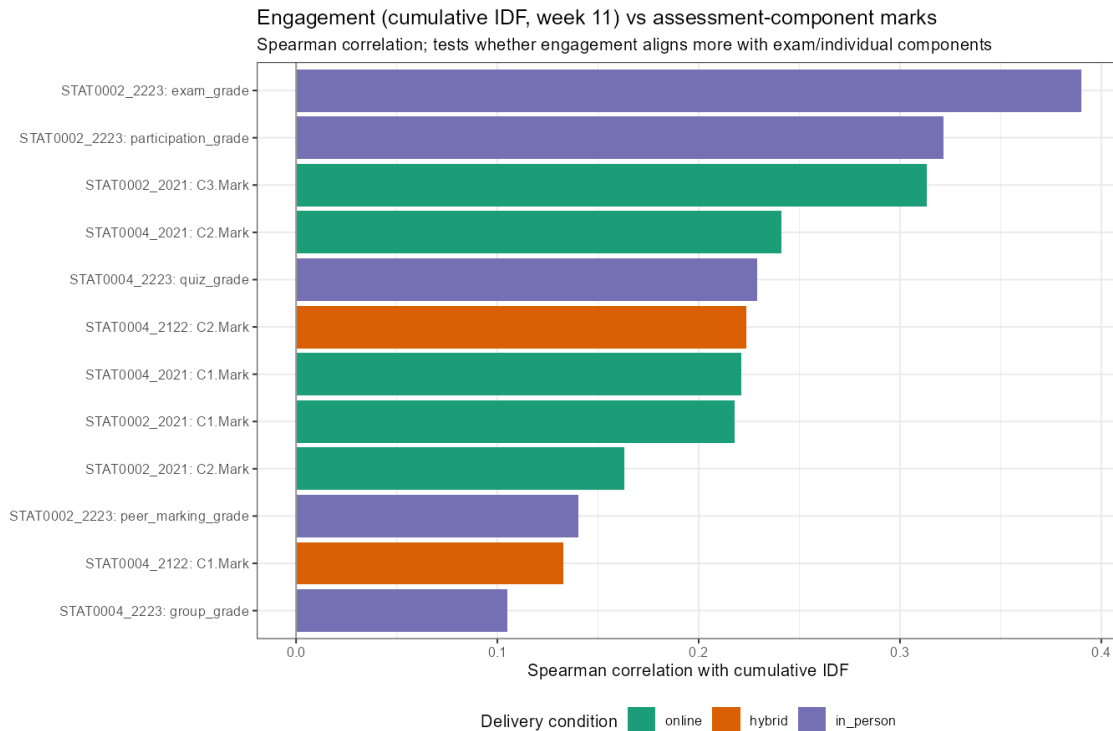


Fig. 11: Engagement–assessment component correlations by cohort.

Timing and regularity features add incremental explanatory power beyond F, I and D in the STAT0002 cohorts, where engagement showed stronger alignment with individual assessment components (Table 10, Figure 12). At week 6, the gain in R^2 is about 0.05 in both STAT0002 deliveries ($p = 0.003$ and $p = 0.006$), but not statistically significant in the STAT0004 deliveries. How regularly students work carries additional diagnostic information in STAT0002, beyond what the core composite already captures.

Table 10: Incremental R^2 from temporal features at week 6.

Cohort	R^2 (F/I/D)	R^2 (full)	ΔR^2	p
STAT0002 online	0.091	0.143	0.052	0.003
STAT0002 in-person	0.156	0.213	0.057	0.006
STAT0004 online	0.089	0.094	0.005	0.939
STAT0004 hybrid	0.069	0.102	0.033	0.424
STAT0004 in-person	0.040	0.094	0.053	0.094

Source: author's analysis of UCL Moodle log data.

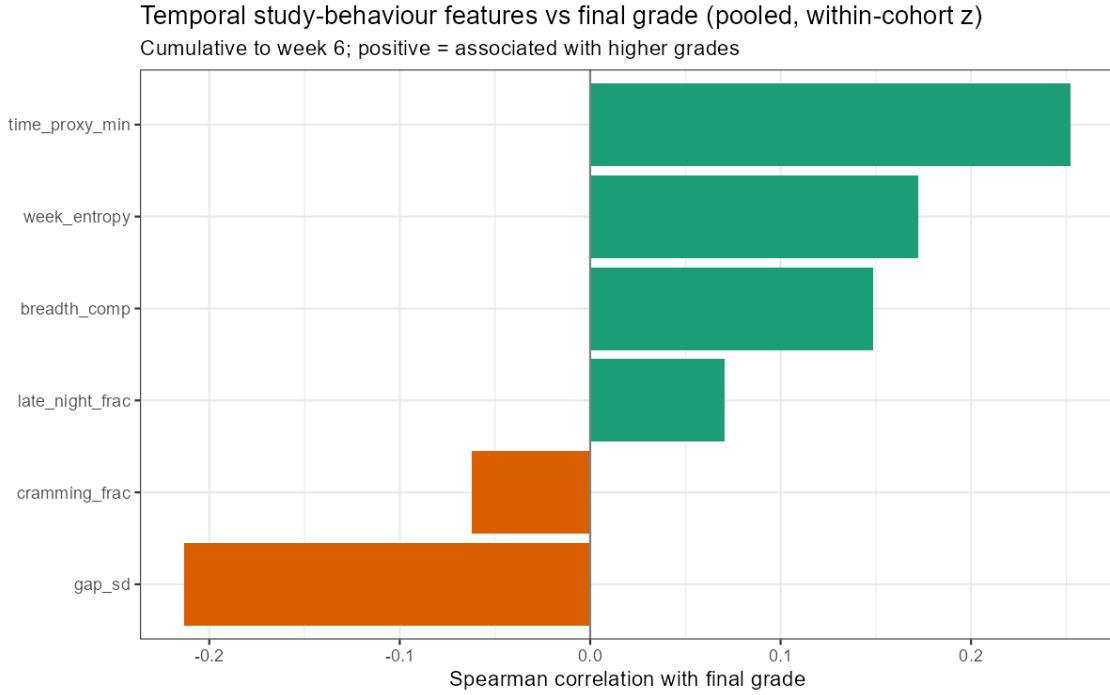


Fig. 12: Temporal and regularity features correlated with final grade.

The diagnostic extensions are not uniform across modules. Temporal regularity adds significant incremental R^2 only in STAT0002, where engagement also aligns more strongly with individually assessed components. In STAT0004, where associations are weaker and group or coursework components carry more weight, the composite and its extensions explain less outcome variance, consistent with greater off-platform study, second-year autonomy, or looser chapter–assessment alignment weakening the Moodle signal rather than invalidating the metric outright.

5.7 Early-warning performance

Although the association is stronger in the lower grade tail, operational flagging still faces modest precision. At the course midpoint, the bottom engagement quintile in STAT0002 identifies roughly 38–47% of students who ultimately score below 50% (AUC around 0.71–0.76), but precision is only

about 15–26%: roughly three-quarters to four-fifths of flagged students may still pass. Figure 13 illustrates this recall–precision trade-off. The metric is therefore useful for identifying students with low *observed* Moodle engagement, not for identifying students who will fail. Appropriate responses are broad, low-cost and supportive rather than intensive or punitive.

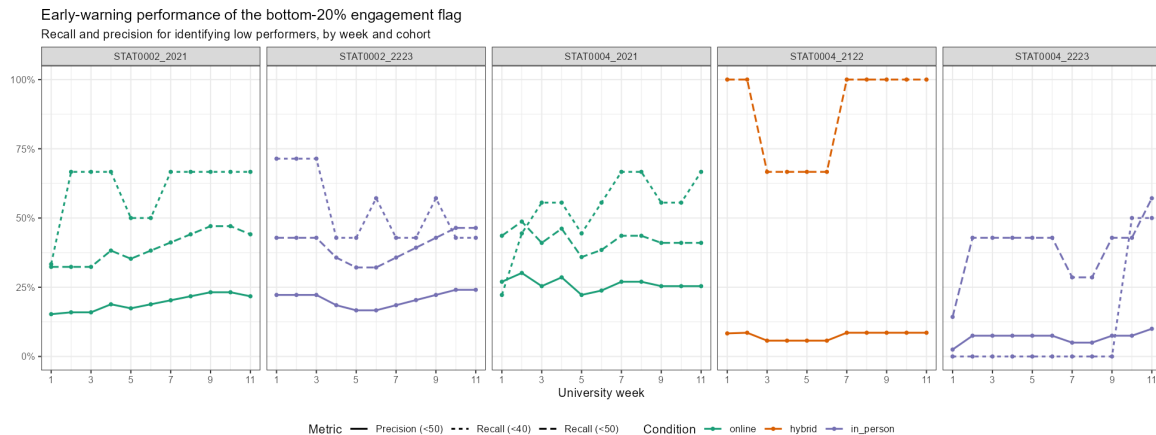


Fig. 13: Recall and precision for the bottom engagement quintile across weeks. Modest precision implies that most flagged students may still achieve passing grades.

5.8 Boundary condition: delivery period and cohort confounding

Because delivery condition is inseparable from cohort and year in this dataset, comparisons across online, hybrid and in-person periods are best treated as a descriptive boundary check rather than a test of delivery effects. Meta-analytic moderator tests do not suggest that delivery label explains between-cohort variation (all $p > 0.87$; Table 2), and weekly correlations appear broadly overlapping across periods (Figure 14). This should not be interpreted as evidence that delivery mode is irrelevant. Module design, assessment structure, cohort composition and Moodle resource organisation may be at least as important as the delivery label itself.

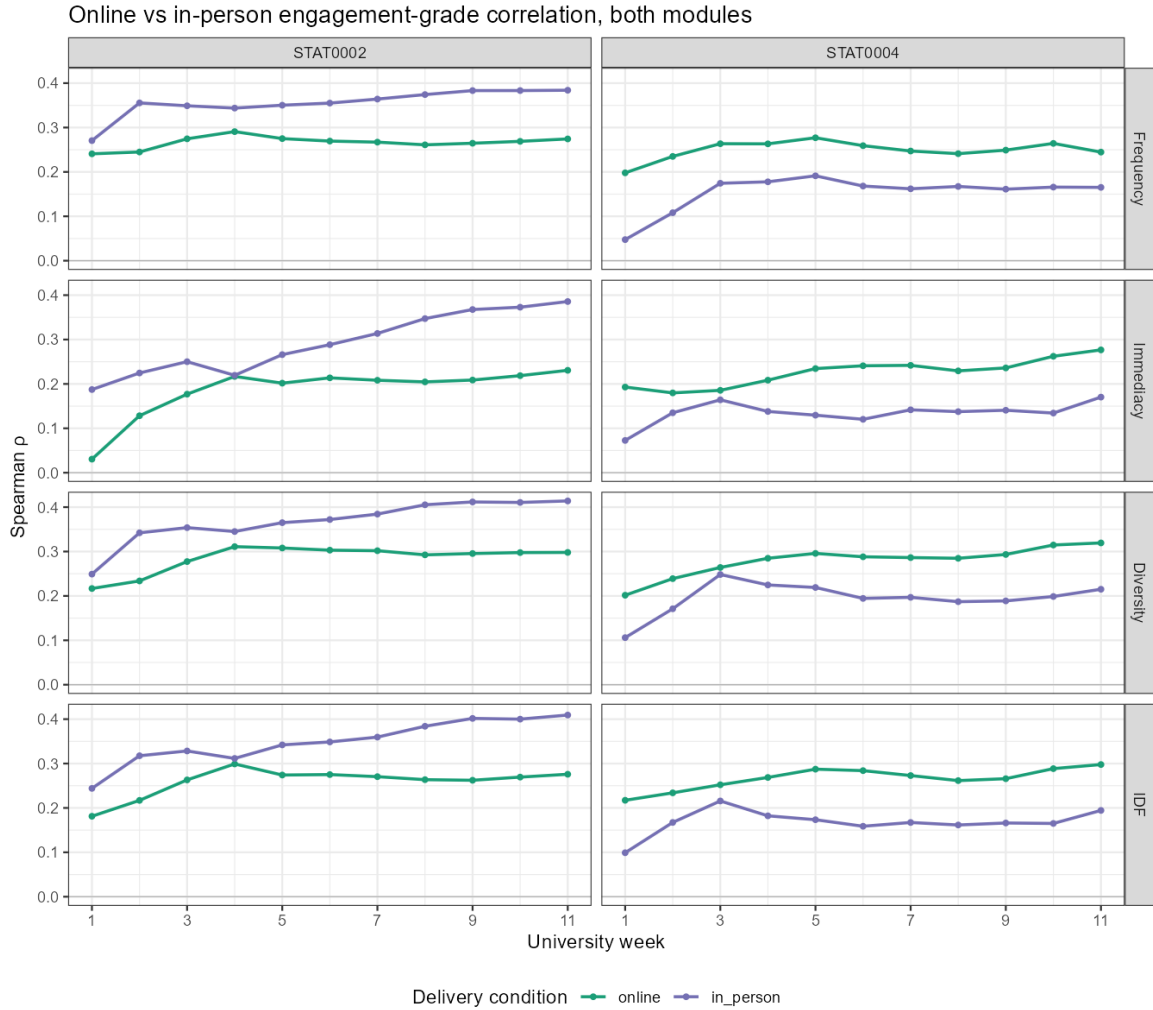


Fig. 14: Weekly engagement–grade correlation by module and delivery period. Comparisons are descriptive; condition is confounded with academic year.

5.9 Robustness checks

Each check addressed a specific threat to the headline conclusions. Programme adjustment tests whether the pooled engagement slope reflects degree-programme composition rather than study behaviour itself; the slope changes by at most 14%, so programme mix is not driving the result. False-discovery-rate correction using the Benjamini–Hochberg procedure guards against spurious weekly associations from repeated testing across weeks and indicators; 205 of 220 tests remain significant. Alternative bounded-outcome specifications address skew and ceiling effects in grade distributions and agree in sign with the main models in all five cohorts. Finally, varying the at-risk engagement threshold examines whether early-warning conclusions depend on an arbitrary quintile cut-off; recall and precision shift in the expected direction without changing the overall interpretation.

6 Discussion

6.1 Interpretation of the component and distributional findings

The central contribution of this report is not that Moodle activity predicts grades perfectly: it does not. Rather, it shows that a transparent, chapter-aligned summary of observed engagement

retains a moderate relationship with final performance in cohorts independent of the original validation. The pooled association ($r = 0.32$) and positive direction in every cohort extend the evidence of Johnston et al. (2025), while also illustrating a central caution in learning analytics: trace data are indicators of observable behaviour, not direct measures of learning, motivation or effort (Winne, 2020; Motz et al., 2019).

Once the overlap among Frequency, Immediacy and Diversity is taken into account, Diversity and Immediacy carry more unique information about final grades than Frequency. This does not imply that session frequency is unimportant. Instead, it suggests that repeated access alone is an incomplete account of productive online study. Breadth of resource use is consistent with curriculum coverage, while prompt first access is consistent with keeping pace with the intended teaching sequence. This interpretation accords with work showing that frequency, immediacy and regularity capture complementary aspects of online engagement (Hoffman et al., 2023; You, 2016). The contrast between similar bivariate correlations and unequal relative-importance shares also demonstrates why component interpretation should not rely only on individual correlations.

The quantile results add a distributional qualification. Engagement is associated with grade throughout the cohort, but the association is larger among students in the lower tail of the grade distribution. This is an empirical finding of the present analysis rather than evidence that increasing Moodle engagement would cause grades to rise by the estimated amount. It nevertheless supports treating low observed engagement as one input to supportive outreach for students who may be academically vulnerable, rather than as a tool for ranking all students with the same confidence.

6.2 Portability, trajectories and diagnostic value

The metric performs differently across the two modules. The stronger engagement–grade associations in STAT0002 than in STAT0004 are plausible given differences in assessment structure, student autonomy and the extent to which study activity occurs outside Moodle. Such contextual variation is expected: the validity and utility of activity logs depend on instructional conditions and course design (Gašević et al., 2016; Motz et al., 2019). It reinforces the need to validate the metric locally rather than assume that a threshold or effect size transfers unchanged to every module.

The extension analyses clarify what is gained by retaining the temporal structure of engagement. Low-steady, mid-variable and high-steady trajectories separate final grades more sharply than a single score at one time point, in line with evidence that longitudinal engagement profiles can reveal meaningful differences in learning outcomes (Saqr and López-Pernas, 2021; Cerezo et al., 2016). Regularity features add information in STAT0002 but not in STAT0004, which is informative rather than a failure: it indicates that temporal patterns should be treated as context-specific diagnostic features. Similarly, stronger correlations with individually assessed components than with peer- or group-marked outcomes are consistent with the fact that group marks do not map cleanly onto individual study behaviour (Almond, 2009).

6.3 Implications for early support

The bottom-engagement flag identifies students with low *observed* Moodle engagement, not students who will fail. With precision of about 15–26% at the course midpoint in STAT0002, most flagged students may still pass. This trade-off is common in early-warning systems and should determine the intensity of any response (Conijn et al., 2017; Akçapınar et al., 2019). Appropriate uses are broad, low-cost actions such as study reminders, non-accusatory check-ins, invitations to office hours, human review, or dashboard-level monitoring of chapters and resources. Engagement should be combined with coursework information, attendance where available, tutor knowledge and student self-report before substantive action.

The component and trajectory results make this approach more informative than a single risk label. A student may be behind the release schedule (low Immediacy), using only a narrow range of materials (low Diversity), engaging irregularly, or following a persistently low trajectory. From week 4 onward, the trajectory analysis can identify a sustained low-engagement pattern with substantially higher precision than the bottom-quintile rule identifies eventual low performance. These distinctions can guide supportive, proportionate outreach. They do not justify punitive messaging, high-stakes ranking, or an assumption that low Moodle activity means low effort.

6.4 Limitations

The findings are associational and their scope is limited. Delivery condition is perfectly confounded with academic year and cohort, so the data cannot identify causal effects of online, hybrid or in-person delivery. The analysis covers two statistics modules and five deliveries; external validity therefore remains to be tested across disciplines, assessment designs and institutional contexts. Moodle logs capture online behavioural engagement only, leaving offline study, motivation, cognitive engagement and accessibility constraints unobserved (Martin and Borup, 2022; Nkomo et al., 2021). Final grades are imperfect outcome measures, especially when group, peer or non-invigilated assessment contributes. Small numbers of low-performing students in some cohorts also make early-warning estimates unstable. Finally, the metric depends on reliable chapter labelling and resource organisation. Any operational deployment should therefore include local validation of data quality, association strength and intervention thresholds.

7 Conclusion

This report validates an existing real-time Moodle engagement metric on new STAT0002 and STAT0004 cohorts spanning online, hybrid and in-person delivery periods. Cumulative engagement is moderately associated with final grades ($r = 0.32$ pooled). Two findings receive particular emphasis. **First**, decomposing engagement into Frequency, Immediacy and Diversity shows that breadth and promptness of resource use outweigh raw session frequency once indicator overlap is removed, a distinction that bivariate correlations alone would not reveal. **Second**, quantile regression shows that engagement matters more for lower-performing students than for the average student, supporting targeted support for at-risk learners rather than uniform prediction across the grade distribution. Trajectory classes, temporal regularity and assessment-component alignment provide additional diagnostic context. Delivery comparisons are descriptive only. The metric should be used for low-cost supportive monitoring with local validation, not high-stakes prediction.

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